

The empirical recurrence for OEIS A237294: recovering a conjecture that is not written down

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Abstract

OEIS A237294 records “Empirical recurrence of order 53”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 248$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 53, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 53 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A237294 is “Number of $(n+1) \times (2+1)$ 0..3 arrays with the maximum plus the minimum minus the upper median minus the lower median of every 2×2 subblock equal.”. It has offset 1 and begins

1128, 7544, 56752, 471588, 4167736, 38817072, 375021656, 3729166264, ...

The entry states:

Empirical recurrence of order 53 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 10:01:48, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 248$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 248$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 496$ exact terms of the model returns order 53 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{53} c_i a(n-i),$$

c_1	53	c_{15}	1338173563718	c_{29}	117367216073692048	c_{43}	−63037230689222656
c_2	−1140	c_{16}	9492154335912	c_{30}	151009106747908592	c_{44}	46001530159235072
c_3	11678	c_{17}	−29305132977789	c_{31}	−296715147143660608	c_{45}	17706446190804992
c_4	−32461	c_{18}	−30085868046881	c_{32}	−126981351065282368	c_{46}	−11694605723500544
c_5	−523404	c_{19}	257489448831288	c_{33}	500434634473985728	c_{47}	−3740691508232192
c_6	5767853	c_{20}	−153066941219678	c_{34}	−3102843776614528	c_{48}	1814496588857344
c_7	−13710453	c_{21}	−1296432716290988	c_{35}	−602255317442269952	c_{49}	551807721930752
c_8	−129817177	c_{22}	2251134779386392	c_{36}	167730836558410752	c_{50}	−139294664556544
c_9	986652334	c_{23}	3646477549029610	c_{37}	531317735875465216	c_{51}	−45382537052160
c_{10}	−690507349	c_{24}	−12715397070080678	c_{38}	−249823966314754048	c_{52}	2196137574400
c_{11}	−18216179105	c_{25}	−2482835007163544	c_{39}	−347998517169991680	c_{53}	939524096000
c_{12}	67711277361	c_{26}	43963203572060840	c_{40}	212478110602682368		
c_{13}	100128525146	c_{27}	−24226847876713368	c_{41}	170543535358017536		
c_{14}	−1160055151465	c_{28}	−100727365559931216	c_{42}	−119891033208619008		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 53, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $53 + 4$ terms removed returns order 53 as well, so $\deg q_{\min} = 53 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $53 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 53 that this sequence satisfies — and that family turns out to have exactly one member.

References

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